D2.2 Gene expression

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d2-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 670–680 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Controlling expression

D2.2.1 Gene expression

pp. 670–671

Gene expression is how the information in a gene affects the phenotype: the gene is **transcribed** into mRNA, the mRNA is **translated** into a polypeptide, and the protein acts, for example as an enzyme. For example, the enzyme tyrosinase is needed to make the pigment melanin.

D2.2.2 Transcription factors

pp. 671–672

Transcription factors are proteins that bind specific base sequences. General factors and RNA polymerase bind the **promoter**, just before the gene. Activators bind **enhancers**, which may be thousands of bases away; the DNA loops so they meet the promoter proteins and raise the rate of transcription. Repressors bind silencers and lower it.

D2.2.3 mRNA breakdown

p. 672

Each mRNA has a poly-A tail that is shortened over time; when it is short enough, nucleases break the mRNA down. In human cells mRNA lasts from minutes to days. Controlling how long it lasts controls how much protein is made from it.

Epigenetics

D2.2.4 Epigenesis

pp. 672–673

Epigenesis is the development of patterns of differentiation in a multicellular organism. Cells switch different genes on and off, so the **phenotype** of each cell changes while the base sequence, the **genotype**, stays the same.

D2.2.5 Genome, transcriptome, proteome

pp. 673–674

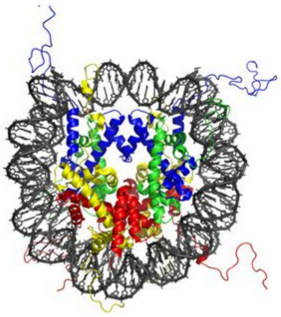
The **genome** is all the genetic information, the same in every cell. The **transcriptome** is the mRNA a cell makes at a given time, and the **proteome** is the proteins it makes; both differ between cells and change over time. No cell expresses all its genes, and the pattern decides how it differentiates.

D2.2.6 Methylation as an epigenetic tag

pp. 674–675

Epigenetic tags: methylation

D2.2.6



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	DNA methylation	Histone methylation
Where	Cytosine in the promoter	Histone tails
Effect	Represses the gene	Can repress or activate

Other tags

Acetylation of histones loosens packing and usually activates genes. Enzymes add and remove tags, so they can change during life.

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Methyl groups added to **cytosine** in a promoter repress transcription of the gene. Methyl groups on amino acids in **histone** tails can repress or activate transcription by changing how tightly DNA is packed and which proteins bind.

D2.2.7 Epigenetic inheritance

pp. 675–676

Methylation patterns are copied when DNA replicates, so daughter cells keep the same pattern of expression. If tags survive meiosis, changes in phenotype can pass to offspring without any change to the base sequence.

D2.2.8 Environmental effects

pp. 676–677

Air pollution (particulate matter, ozone, nitrogen oxides) is linked to changes in DNA methylation, for example in genes for immune responses, and to asthma. In agouti mice, a mother's diet changes methylation and the coat colour and weight of her offspring. Much human evidence is correlational.

D2.2.9 Resetting tags in gametes

pp. 677–679

Most tags are removed in eggs and sperm and in the early embryo, so the zygote can form every cell type. Some survive: in **imprinted** genes one parent's copy is silenced. Ligers (lion father, tiger mother) grow very large, probably because growth-promoting imprints from the lion are not balanced by growth-limiting imprints from the tiger; tigons do not.

D2.2.10 Twin studies

pp. 679–680

Monozygotic twins have identical genomes, so differences between them come from the environment. Their methylation patterns are similar when young and differ more with age, especially if they live apart.

D2.2.11 External signals

p. 680

In *E. coli*, **lactose** binds a repressor so that it leaves the DNA, and the genes for lactose uptake and breakdown are transcribed only when lactose is present. A steroid hormone such as **oestradiol** enters target cells, binds a receptor, and the complex acts as a transcription factor.

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